Task 4: Order Management using Joins

Aim: To design and develop an order management system using SQL joins and subqueries to display customer order history, identify the highest value order, and determine the most active customer.

Algorithm:

1. Start the program
2. Connect to the database
3. Retrieve data from Customers, Orders, and Products tables
4. Apply JOIN to combine data from all tables
5. Calculate total order value (price × quantity)
6. Sort data using ORDER BY
7. Find highest value order using subquery (MAX)
8. Find most active customer using GROUP BY and COUNT
9. Display results on the web page
10. Stop the program

Program:

Orders.php:

```
<?php

include 'db.php';

/* ==================================

   1. CUSTOMER ORDER HISTORY (JOIN)

================================== */

$sql = "SELECT c.name, p.product_name, p.price, o.quantity, o.order_date,

        (p.price * o.quantity) AS total

        FROM orders o

        JOIN customers c ON o.customer_id = c.customer_id

        JOIN products p ON o.product_id = p.product_id

        ORDER BY o.order_date DESC";

$result = $conn->query($sql);

/* ==================================

   2. HIGHEST VALUE ORDER (SUBQUERY)

================================== */

$highest_sql = "SELECT MAX(p.price * o.quantity) AS max_order
```

```
               FROM orders o
               JOIN products p ON o.product_id = p.product_id";
$highest_result = $conn->query($highest_sql);
$highest_row = $highest_result->fetch_assoc();
/* ================================
   3. MOST ACTIVE CUSTOMER
================================= */
$active_sql = "SELECT c.name, COUNT(*) AS total_orders
              FROM orders o
              JOIN customers c ON o.customer_id = c.customer_id
              GROUP BY c.customer_id
              ORDER BY total_orders DESC
              LIMIT 1";
$active_result = $conn->query($active_sql);
$active_row = $active_result->fetch_assoc();
?>
<!DOCTYPE html>
<html>
<head>
    <title>Order Management System</title>
    <style>
        body { font-family: Arial; background: #f2f2f2; }
        .container {
            width: 85%;
            margin: auto;
            background: white;
            padding: 20px;
        }
        table {
```

```
            width: 100%;
            border-collapse: collapse;
            margin-top: 20px;
        }
        th, td {
            padding: 10px;
            border: 1px solid black;
            text-align: center;
        }
        h2 {
            color: blue;
        }
        .highlight {
            background: #e6f7ff;
            padding: 10px;
            margin-top: 15px;
        }
    </style>
</head>
<body>
<div class="container">

    <h2>Customer Order History</h2>
    <table>
        <tr>
            <th>Customer</th>
            <th>Product</th>
            <th>Price</th>
            <th>Quantity</th>
```

```
            <th>Total</th>
            <th>Date</th>
        </tr>
        <?php while($row = $result->fetch_assoc()) { ?>
        <tr>
            <td><?php echo $row['name']; ?></td>
            <td><?php echo $row['product_name']; ?></td>
            <td><?php echo $row['price']; ?></td>
            <td><?php echo $row['quantity']; ?></td>
            <td><?php echo $row['total']; ?></td>
            <td><?php echo $row['order_date']; ?></td>
        </tr>
        <?php } ?>
    </table>
    <div class="highlight">
        <h2>Highest Value Order</h2>
        <p>₹ <?php echo $highest_row['max_order']; ?></p>
    </div>
    <div class="highlight">
        <h2>Most Active Customer</h2>
        <p>
            <?php echo $active_row['name']; ?>
            (<?php echo $active_row['total_orders']; ?> orders)
        </p>
    </div>
</div>
</body>
</html>
```

Database:

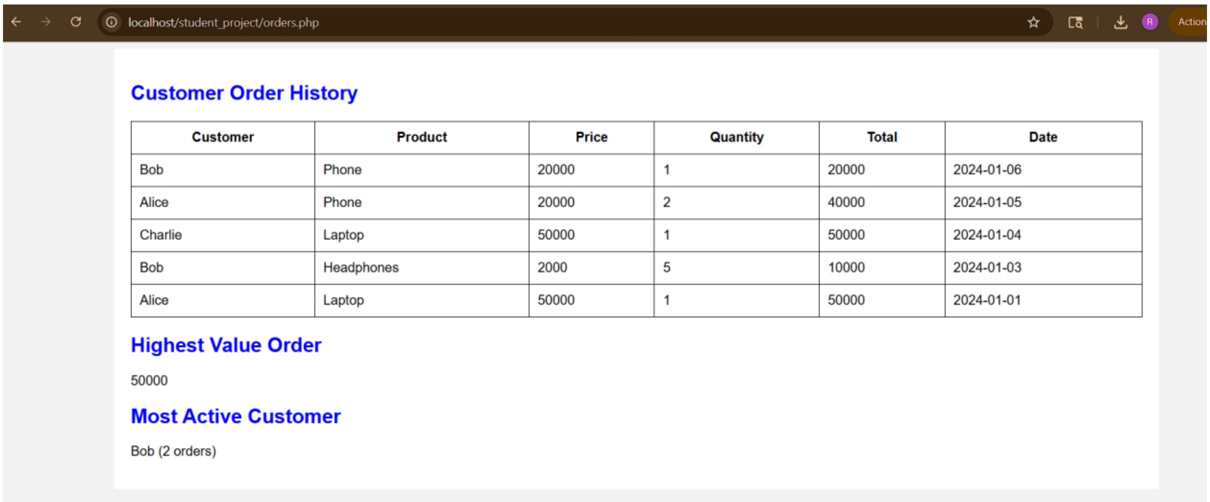
```sql
CREATE DATABASE order_db;

USE order_db;

CREATE TABLE customers (
    customer_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100)
);

CREATE TABLE products (
    product_id INT AUTO_INCREMENT PRIMARY KEY,
    product_name VARCHAR(100),
    price INT
);

CREATE TABLE orders (
    order_id INT AUTO_INCREMENT PRIMARY KEY,
    customer_id INT,
    product_id INT,
    quantity INT,
    order_date DATE
);

-- Sample Data

INSERT INTO customers (name) VALUES
('Alice'), ('Bob'), ('Charlie');

INSERT INTO products (product_name, price) VALUES
('Laptop', 50000),
('Phone', 20000),
('Headphones', 2000);

INSERT INTO orders (customer_id, product_id, quantity, order_date) VALUES
(1,1,1,'2024-01-01'),
(1,2,2,'2024-01-05'),
```

(2,3,5,'2024-01-03'),

(3,1,1,'2024-01-04'),

(2,2,1,'2024-01-06');

Output:



Customer Order History

Customer	Product	Price	Quantity	Total	Date
Bob	Phone	20000	1	20000	2024-01-06
Alice	Phone	20000	2	40000	2024-01-05
Charlie	Laptop	50000	1	50000	2024-01-04
Bob	Headphones	2000	5	10000	2024-01-03
Alice	Laptop	50000	1	50000	2024-01-01

Highest Value Order

50000

Most Active Customer

Bob (2 orders)

Result: Thus, the order management using joins in sql has been successfully completed and output generated successfully.